IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART II: MEng, BEng and ACGI

Corrected Copy

MATHEMATICS 2A (E-STREAM AND I-STREAM)

Thursday, 23 May 2:00 pm

Time allowed: 1:30 hours

There are FOUR questions on this paper.

Answer ALL questions

NO CALCULATORS ALLOWED. Table of Laplace transforms included

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : B.K.W. Ee

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EE2-08A MATHEMATICS

1. a) Show that the function $v(x,y) = e^{-y}\sin x - e^{x}\cos y$ satisfies Laplace's equation and integrate the Cauchy-Riemann equations to find its harmonic conjugate $u(x,y)$. [5]

   b) Hence obtain $f(z) = u(x,y) + iv(x,y)$ where f is an analytic function of $z = x + iy$, simplifying as much as possible. [4]

2. a) The complex function

$$F(z) = \frac{e^{imz}}{(z^2+1)(z^2+4)}$$

has four simple poles. Find the residues at the poles lying in the upper half of the complex plane. [5]

   b) Consider the contour integral $I = \oint_C \frac{e^{imz}}{(z^2+1)(z^2+4)}\,dz$,

where the closed contour C consists of a semi-circle in the complex upper half-plane, taken in the anti-clockwise sense, and $m > 0$.

Using the result from (a), obtain

$$\int_{-\infty}^{\infty} \frac{\cos mx}{(x^2+1)(x^2+4)}\,dx,$$

explaining carefully your evaluation of I, and your use of Cauchy's Residue Theorem and Jordan's lemma. [8]

Recall that the residue of a complex function $F(z)$ at a pole $z = a$ of multiplicity m is given by the expression

$$\lim_{z\to a}\left\{\frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\left[(z-a)^m F(z)\right]\right\}.$$

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3. a) The complex function

$$f(z) = \frac{g(z)}{h(z)},$$

has a simple pole at $z = a$ if $h(z)$ has a simple zero at $z = a$, while $g(a) \neq 0$. In that case, use a Taylor expansion of $h(z)$ to show that the residue of $f(z)$ at $z = a$ is [5]

$$\text{Res}\,[f(z)]_{z=a} = \frac{g(a)}{h'(a)}.$$

b) Show also, that if $g(z) = 1$, a constant, and $f(z)$ has a double pole at $z = a$, then the residue is [7]

$$\text{Res}\,[f(z)]_{z=a} = -\frac{2h'''(a)}{3[h''(a)]^2}.$$

4. a) A function $f(t)$ has Laplace transform $\mathscr{L}[f(t)]$. Prove the second shift-theorem, that for a positive constant a:

$$\mathscr{L}[H(t-a)f(t-a)] = e^{-as}\mathscr{L}[f(t)],$$

where $H(t)$ is the unit-step, or Heaviside function. [6]

b) Consider the following second-order ODE

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x = \delta(t-2)$$

satisfying the initial conditions $x(0) = \frac{dx}{dt}(0) = 0$, where $\delta(t)$ is the unit-impulse, or Dirac delta-function.

Use the Laplace convolution theorem to show that $x(t)$ satisfies

$$x(t) = 0 \quad \text{for} \quad 0 \leq t \leq 2,$$

and find the form of $x(t)$ for $t > 2$. [10]

Table of Laplace transforms

$f(t)$	$\mathcal{L}\{f(t)\} \equiv F(s)$
A	$\frac{A}{s}$, $\quad \mathrm{Re}(s) > 0$
e^{at}	$\frac{1}{s-a}$, $\quad \mathrm{Re}(s) > a$
$t^n, \quad n = 1, 2 \ldots$	$\frac{n!}{s^{n+1}}$, $\quad \mathrm{Re}(s) > 0$
$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$, $\quad \mathrm{Re}(s) > 0$
$\cos \omega t$	$\frac{s}{s^2+\omega^2}$, $\quad \mathrm{Re}(s) > 0$
$e^{at} f(t)$	$F(s-a)$
$t^n f(t)$	$(-1)^n \frac{d^n F}{ds^n}$
$\frac{df}{dt}$	$sF(s) - f(0)$
$\frac{d^2 f}{dt^2}$	$s^2 F(s) - sf(0) - \frac{df}{dt}(0)$
$H(t-a)$	$\frac{e^{-as}}{s}$
$\delta(t-a)$	$e^{-as}, \quad a > 0$
$f(t-a)H(t-a)$	$e^{-as}F(s)$
$\int_0^t f(u)g(t-u)\,du$	$F(s)G(s)$